



W3C VCWG TF Vocabularies for DPP

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June 15, 2026

Agenda for June 15 meeting

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- Recap of June 3 F2F meeting
 - Interest for DPP vocab
 - Agreeing on what we want to achieve in this group/next steps

Digital Product Passport in a Nutshell

A **Digital Product Passport (DPP)** is an **online** collection of **machine-readable** product data accessed through a **data carrier**.

DPPs are being currently being introduced both as **mandatory** and **voluntary** instruments all over the world, for different reasons:

- Sustainability & Market Surveillance (EU) – **Mandatory**
- Innovation (all of industry)
- Traceability of ESG information (UN)
- Customs clearance efficiency (USA)
- Consumer protection (Vietnam)
- Digitalisation/efficiency (China)
- Etc.



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- DPP schemes/standards are a highly political issue.
- Both JSON & JSON-LD will be used for DPPs.
- Some DPPs will be issued as VCs.
- Each DPP scheme will likely use different vocabularies.
- The EU Commission will likely create vocabularies for the EU DPP.
- DPPs and organizational identities are closely related.

The problem I am trying to solve : Reusing product data from any DPP issued by any nation/organization.
→ Combining data from different DPP flavors.
→ Linking different DPPs & product-related attestations according from different DPP flavors.

2 major challenges for the success of DPPs



Companies may be forced to place multiple data carriers on their products to satisfy differing regulatory requirements in different markets.



Conflict of law: “Regulator A” could impose mandatory requirements on “Regulator B”.



Interest for DPP Vocab

- Editor: Ivo Ladenius, GS1 NL.
- Ivan Herman, W3C
- Ted Guild (Geotab), Covesa
- Eva Blomqvist - Linköping University
- Susanne Guth-Orlowski, 4theRecord
- Steve Capell, UN/CEFACT
- Robert David, Graphwise
- Wajih Rehman, Fabacus
- Michael Shea, The Aeolian
- Amir Hameed Mir, Sirraya Labs.
- Rigo Wenning
- Ingo (Spherity)
- Monika Nowicki (interest for EBW focus, recognized entities)
- Seth (EC/DG AGRI)
- ...

Recap of June 3 F2F meeting discussions

- Is the VCWG the right place to discuss design of use-case specific vocabularies?
 - No (in principle), because the use case representatives are not in VCWG.
 - But right now, many VC user groups don't know where else to go... And W3C is a good place to meet international adopters of VC-tech. And many vocabularies don't have the appropriate "homes" yet, so people are asking for (temporary) W3C namespaces.
 - How can VCWG support such communities wishing to develop VC-based use cases, especially international/cross-border ones?
 - VCWG:
 - Providing a test suite for these (real-world) VC use cases vocabularies, to prove they are W3C VCDM-compatible.
 - Implementation of a VC-vocabulary registry? (implications of endorsement by W3C? Needs AC approval) Qualification criteria for registry ? Registry metadata? Registry metrics. Vocabulary use tracking ?
 - DPP TF
 - The DPP TF should prove that the VCDM VC is appropriate for DPP application vocabs.
 - Creating a guidance doc for VCDM use for DPP issuance & combinability.
 - W3C interest group for use case vocabulary designers?
 - Vocabulary mapping exercise ? (Note that Carolyn and Eva are already working on a publication on this topic.)
 - **POSTPONED** work for this TF:
 - Is there value for "**Yet another DPP vocab**"? A W3C vocab for data integration ? move to ISO in a later stage ?
 - Methodology
 - Thoughts/discussion points should be expressed as github issues. → e.g. Why a DPP should be issued as a VC ?

- Why use a VC for DPPs at all ? This needs to be better understood in the EU DPP system context.
- Applicability of VCDM model to DPPs (EU/others)
 - Susanne's question about credential subject...
 - A test-suite for DPPs based on VCDM ?
 - A DPP-playground like environment
 - An EU-DPP implementation based on VCDM. (interest from European Commission)
- Definition of DPP “Use cases”:
 - Single-standard “use cases”:
 - Issuing an EU-compliant DPP
 - Issuing an UNTP-compliant DPP
 - Issuing an XYZ-compliant DPP
 - Multi-standard “use cases” for a W3C core DPP vocabulary (a future ISO standard?) :
 - Linking DPPs (possibly of different flavors) : a product with a DPP can have a component/material that has a DPP.
 - Linking product-related claims using different standards e.g. EU DPP containing a link to an UNTP claim.
 - Retrieving and combining data from DPPs from multiple standards.

Why issue a DPP as a VC ?

- A verifiable credential (VC) is a mechanism for issuing authenticated, revokable and portable data in a tamper-proof way and for providing privacy-preserving access to this data by a third party (the “verifier” or the “relying party”). Thus, VCs are useful tools for encapsulating product-related claims: e.g. providing copyright proofs for merchandised products, ESG performance certifications, declarations of conformity, customs clearance proofs, etc., that might be accessed from a DPP.
- Different questions arise when considering issuing the DPP itself as a VC. In the normal case of VC use, a credential is issued by the “issuing party” to another party, the “holder” in recognition by the “issuer” of claims concerning the “holder”. However, differently from most applications of VCs, where the “issuer” and “holder” are distinct parties and the “verifier” retrieves the data from the “holder” and only checks with the “issuer” that the credential is still valid, the default situation for a DPP (in particular for the European DPP) is that the “issuer” and “holder” are the same organisation, meaning that the DPP data is retrieved via a secure HTTPS request directly from the issuing organisation, a.k.a. the “economic operator”. HTTPS provides the minimum requirements for authentication and integrity (assuming the link used to retrieve the data is trusted).
- The following list provides a number of use cases where VCs provide additional capabilities with respect to HTTPS.
 1. **Identifying the issuer of the DPP:** In many trade and logistics supply-chains, the DPP data may be stored and used locally by different “holders”. Thus, the server hosting the DPP may provide no information on the actual issuer of the data.
 2. **DPP revocation:** As economic operator are liable for the accuracy of the DPP data, DPP revocation is a useful feature in case an error has been detected and a new DPP must be issued. Revocation can also be useful for products with “use until” constraints.
 3. **Proving data authenticity:** In a world where it is very easy to generate fake product data, an organisation making use of DPP data for their own commercial operations (e.g. a professional purchasing company) can limit their liability if they are able to provide proofs that they are indeed using authentic data.
 4. **Privacy:** The default HTTPS request provides no privacy for a party retrieving DPP data, e.g. a consumer. If the DPP is “held” by a third party, e.g. a retail store, the DPP copy held by the third party can be accessed and authenticated by the consumer instead of the version held by the issuing economic operator. This protects the privacy of the consumer. For example, if the DPP is for a medication, access to the DPP can reveal information about the user’s medical condition.
 5. **Timestamps:** The timestamp inherent to the VC issuing helps improving the DPP data quality and supports liabilities related to claims.
 6. **Integrity over time:** Differently from HTTPS which provides integrity during data transmission, the VC’s timestamps also supports DPP integrity, in the sense of absence of change, over time.